Introductory Exam Notes

Jonathan Ma

October 29, 2026

Contents

I Linear Algebra **6**

1 Vectors & Analytic Geometry of Space **7**
1.1 Coordinate Systems in n-space . . . 7
1.2 n-Vectors . . . 7
1.3 Linear Dependence of n-vectors . . . 7
1.4 Length & Inner Product of n-vectors . . . 7
1.5 Outer/Cross Product of 3-vectors . . . 7

2 Finite-Dimensional Vector Spaces **7**
2.1 n-dimensional Vector Spaces Over $\mathbb{R}$ and $\mathbb{C}$ . . . 7
2.2 Linear Dependence & Generators . . . 7
2.3 Simultaneous Linear Equation, Gaussian elimination . . . 7
2.4 Bases & Dimensions . . . 7
2.5 Subspaces . . . 7
2.6 Inner Products; Schwarz Inequality . . . 7
2.7 Orthogonal Bases; Orthogonal Complements; Projections . . . 7
2.8 Dual Spaces . . . 7
2.9 Lines, Hyperplanes, Coordinate systems . . . 7
2.10 Distance; Polarization Identity . . . 7

3 Linear Transformations and Matrices **7**
3.1 Linear Transformations; Elementary Properties (Image/Nullspace) . . . 7
3.2 Addition, Composition, Scalar Multiplication . . . 7
3.3 Matrices & Linear Transformations . . . 7
3.4 Nonsingular Linear Transformations; Inverses . . . 7
3.5 Changes of Bases in Vector Spaces . . . 7
3.6 Similarity of Matrices . . . 7

3.7 Special Types of Square Matrices (Symmetric, Orthogonal, Triangular etc.) 7
3.8 Elementary Matrices 7
3.9 Rank of a Matrix 7

4 Determinants **7**
4.1 Definition & Elementary Properties 7
4.2 Permutations & Uniqueness 7
4.3 Minors; Cofactors; Evaluation of Determinants 7
4.4 Applications (Dependence of Vectors; Volume of a Parallelopiped; Linear Equations) 7
4.5 Determinants of Products & Inverses 7
4.6 Determinants of Special Types of Matrices 7

5 Bilinear and Quadratic Forms **7**
5.1 Bilinear Mappings & Forms 7
5.2 Quadratic Forms; Polarization 7
5.3 Equivalence of Quadratic Forms; Congruence of Matrices 7
5.4 Geometric Applications 7

6 Eigenvalues and Eigenvectors **7**
6.1 Definitions 7
6.2 Similarity & Diagonal Matrices 7
6.3 Orthogonal Reduction of Symmetric Matrices 7
6.4 Eigenvalues of Special Types of Matrices (Unitary, Hermitian, etc.) 7
6.5 Normal Matrices & Spectral Theorem 7
6.6 Singular Values 7

II Probability **7**

7 Combinatorial Analysis **7**
7.1 Inclusion-Exclusion Principle 7
7.2 Permutations 7
7.3 Combinations 7
7.4 Multinomial Coefficients 7

8 Axioms of Probability **7**
8.1 Sample spaces and events 7
8.2 Axioms of Probability 7
8.3 The uniform model 7
8.4 Probability as a continuous set function 7

9 Conditional probability and independence **7**
9.1 Conditional probabilities . . . 7
9.2 Bayes' theorem . . . 7
9.3 Independent events . . . 7

10 Random variables and their distributions **7**
10.1 Random variables . . . 7
10.2 Discrete random variables, Indicator functions . . . 7
10.3 Expected value . . . 7
10.4 Expectation of a function of a random variable . . . 7
10.5 Variance . . . 7
10.6 Distribution function . . . 7
10.7 Probability mass function . . . 7
10.8 Probability density function . . . 7

11 Discrete univariate distributions **7**
11.1 Bernoulli and binomial distributions . . . 7
11.2 Geometric and negative binomial distributions . . . 7
11.3 Poisson distributions . . . 7
11.4 Hypergeometric distributions . . . 7

12 Continuous univariate distributions **7**
12.1 Uniform distributions . . . 7
12.2 Normal distributions . . . 7
12.3 Exponential distributions . . . 7
12.4 Gamma distributions . . . 7
12.5 Cauchy distributions . . . 7
12.6 Change of variables . . . 7

13 Multivariate distributions **7**
13.1 Independent random variables . . . 7
13.2 Convolution and sums of independent random variables . . . 7
13.3 Conditional distributions . . . 7
13.4 Change of variables . . . 7
13.5 Multivariate normal distributions . . . 7

14 Properties of expectation and related matters **7**
14.1 Linearity of expectation . . . 7
14.2 Covariance . . . 7
14.3 Variance of a sum . . . 7
14.4 Correlation . . . 7

14.5 Conditional expectation and the law of total expectation . . . 7
14.6 Conditional variance and the law of total variance . . . 7
14.7 Conditional covariance and the law of total covariance . . . 7
14.8 Conditional expectation and prediction . . . 7
14.9 Moment generating functions . . . 7

15 Inequalities and limit theorems **7**
15.1 Chebychev's and Markov's inequalities . . . 7
15.2 Cauchy-Schwarz inequality . . . 7
15.3 Jensen's inequality . . . 7
15.4 Weak and strong laws of large numbers . . . 7
15.5 Central limit theorem . . . 7

III Real Analysis **7**

16 Basic set theory **7**
16.1 Finiteness, countability, and uncountability . . . 7
16.2 Ordered fields . . . 7
16.3 Cauchy sequences . . . 7
16.4 Construction of the real numbers . . . 7
16.5 Completeness of the real numbers . . . 7
16.6 Infimum and supremum . . . 7
16.7 Liminf and limsup of sequences . . . 7
16.8 Limit points of sets, and sequences . . . 7
16.9 Norms, inner products, and metrics . . . 7
16.10 Open sets and closed sets . . . 7
16.11 Closure and boundary of a set . . . 7
16.12 Compact sets . . . 7
16.13 Bolzano–Weierstrass theorem . . . 7
16.14 Heine–Borel theorem in Euclidean space . . . 7
16.15 Intersection of nested compact sets . . . 7

17 Continuous mappings **7**
17.1 Sums, products, compositions of continuous mappings . . . 7
17.2 Path-connected and connected sets . . . 7
17.3 Images of compact and connected sets . . . 7
17.4 Extreme value theorem . . . 7
17.5 Intermediate value theorem . . . 7
17.6 Uniform continuity . . . 7

18 Differentiability **7**
18.1 Chain rule . . . 7
18.2 Product rule . . . 7
18.3 Mean value theorem . . . 7
18.4 Taylor's theorem . . . 7
18.5 Derivative matrix . . . 7
18.6 Continuity of differentiable mappings . . . 7
18.7 Differentiable paths . . . 7
18.8 Directional derivatives . . . 7
18.9 Inverse function theorem . . . 7
18.10 Implicit function theorem . . . 7
18.11 Maxima and minima . . . 7
18.12 Constrained extrema and Lagrange multipliers . . . 7

19 Integration **7**
19.1 Existence and elementary properties of the Riemann integral . . . 7
19.2 Fundamental theorems of calculus . . . 7
19.3 Differentiation of limits of integration . . . 7
19.4 Interchange of order of integration (Fubini's theorem) . . . 7
19.5 Differentiation of integrals with respect to a parameter in the integrand . . . 7
19.6 Change of variables theorem . . . 7
19.7 Polar, spherical, and cylindrical coordinates . . . 7
19.8 Improper integrals . . . 7

20 Sequences and series **7**
20.1 Numerical sequence and series . . . 7
20.2 Pointwise and uniform convergence of sequence of functions . . . 7
20.3 The Weierstrass M-test . . . 7
20.4 Differentiation and integration of series . . . 7
20.5 Elementary functions (exponential, trigonometric, hyperbolic) . . . 7
20.6 Contraction mapping theorem . . . 7
20.7 The Stone-Weierstrass theorem . . . 7
20.8 Dirichlet and Abel tests . . . 7
20.9 Power series and Cesàro and Abel summability . . . 7

Part I
Linear Algebra

1 Vectors & Analytic Geometry of Space

1.1 Coordinate Systems in n-space

1.2 n-Vectors

1.3 Linear Dependence of n-vectors

1.4 Length & Inner Product of n-vectors

1.5 Outer/Cross Product of 3-vectors

2 Finite-Dimensional Vector Spaces

2.1 n-dimensional Vector Spaces Over $\mathbb{R}$ and $\mathbb{C}$

2.2 Linear Dependence & Generators

2.3 Simultaneous Linear Equation, Gaussian elimination

2.4 Bases & Dimensions

2.5 Subspaces

2.6 Inner Products; Schwarz Inequality

2.7 Orthogonal Bases; Orthogonal Complements; Projections

2.8 Dual Spaces

2.9 Lines, Hyperplanes, Coordinate systems

2.10 Distance; Polarization Identity

3 Linear Transformations and Matrices

3.1 Linear Transformations; Elementary Properties (Image/Nullspace)

3.2 Addition, Composition, Scalar Multiplication

3.3 Matrices & Linear Transformations

3.4 Nonsingular Linear Transformations; Inverses

3.5 Changes of Bases in Vector Spaces

3.6 Similarity of Matrices

3.7 Special Types of Square Matrices (Symmetric, Orthogonal, Triangular etc.)

3.8 Elementary Matrices

3.9 Rank of a Matrix